# The Problem

In dataset, "Other" in category-10 has 72,581 samples, far more than the other classes (e.g., POLITICS has 28,481).

The “Other” class itself is a mix of 33 smaller subcategories (like SPORTS, ARTS, ENVIRONMENT, etc.), each with different numbers of samples.

Without balancing, the model could overfit on “Other” and ignore smaller classes

# Balancing Approach

To balance the ‘OTHER’ category itself also make the class as all almost balanced with other classes

- **Determine the target sample size** by taking the minimum sample count across all categories inside ‘OTHER’ (n = 811).
- **Downsample each subcategory within ‘Other’** to match this target, preserving smaller subcategories that already have fewer samples.
- **Combine the balanced ‘Other’ subcategories with the remaining non-Other categories**, keeping their original distribution.

This approach results in a more balanced dataset, reducing the risk of bias toward the "Other" category and improving model generalization across all classes.

# 1. Experimental Setup

## Feature Representations

representations were evaluated:

- **Bag of Words (BoW)** using CountVectorizer
- **Part-of-Speech (POS) tags**
- **Named Entity Recognition (NER)**
- **Combined features**:
  - BoW + POS
  - POS + NER
- **NER words only** (entities extracted as tokens)

## Text Preprocessing Variants

- Stemming
- Lemmatization
- Stopwords:
  - Kept
  - Removed

## Models Evaluated

- Support Vector Machine (SVC)
- Multinomial Naïve Bayes
- Multi-Layer Perceptron (MLP / Neural Network)
- Hidden Markov Model (HMM)

## Dimensionality Reduction

- **TruncatedSVD** was applied **only for MLP and HMM**
- Purpose:
  - Reduce memory usage
  - lighten sparsity of high-dimensional BoW/POS/NER matrices
  - Enable HMM training on continuous features

# 2. Results Summary

### Best Accuracy per representation

| Model \ Features | BoW | POS | NER | BoW + POS | POS + NER |
|---|---|---|---|---|---|
| **SVM (SVC)** | 0.6485 | 0.1403 | 0.1569 | 0.6113 | 0.2047 |
| **Multinomial NB** | 0.6146 | 0.2817 | 0.2522 | 0.6137 | 0.3473 |
| **MLP (Neural Network)** | **0.6662** | **0.3800** | **0.3003** | **0.6638** | **0.4071** |
| **HMM** | 0.0730 | 0.0842 | 0.0393 | 0.0889 | 0.0651 |

# 3. Detailed Observations by Feature Type

### BoW-Based Models

- **MLP achieved the highest overall performance (0.6662)** using BoW features.
- SVM and Naive Bayes produced competitive and stable results.
- BoW remains the most informative representation for this dataset.

### POS Features

- Performance dropped significantly for all models.
- Best result:
  - **MLP_POS = 0.38**
- POS tags alone lack semantic richness for classification.

### NER Features

- Very weak standalone performance.
- Best result:
  - **MLP_NER = 0.3003**
- NER captures entities but ignores contextual information.

### NER Words Only

- Identical performance to NER features.
- Indicates entity surface forms alone are insufficient.

### Combined Features (POS + NER, BoW + POS)

- Combining syntactic/semantic features improves performance slightly:
  - **MLP_BoW_POS = 0.6638**
  - **MLP_POS_NER = 0.4071**
- However, gains are marginal compared to pure BoW.

# 4. Model-Specific Analysis

## Support Vector Machine (SVM)

- Strong and consistent on BoW features.
- Highly sensitive to feature quality:
  - Performs poorly on POS and NER.
- Stable performance without heavy dependence on preprocessing variants.

## Multinomial Naive Bayes

- Performs well on count-based features (BoW).
- Performance drops sharply on sparse or low-frequency features (POS/NER).
- Assumption of feature independence limits effectiveness on structured features.

## MLP (Neural Network)

- **Best overall model on features**.
- Benefits from:
  - Rich lexical information
  - Reduced dimensionality via SVD
- Sensitive to noisy and sparse inputs (POS, NER).

## Hidden Markov Model (HMM)

- Worst-performing model across all feature types.
- Best accuracy: ~0.089
- Reasons:
  - HMM assumes sequential dependencies
  - BoW, POS, and NER are unordered representations

# 5. Data Preparation Pipeline (Explanation)

### Step 1: Text Cleaning

- Lowercasing
- Removal of punctuation and non-alphabetic tokens
- Token normalization

### Step 2: Preprocessing

- **Stemming** or **Lemmatization**
- Stopwords optionally removed
- POS tagging using a parser
- NER extraction using named entity recognition

### Step 3: Feature Construction

- **BoW**: word frequency vectors
- **POS**: tag frequency vectors
- **NER**: entity tag frequencies or entity words
- Combined features were created via feature concatenation

### Step 4: Vectorization

- CountVectorizer used for all sparse representations
- Test data transformed using the same vocabulary

### Step 5: Dimensionality Reduction (MLP & HMM only)

- TruncatedSVD applied to sparse matrices
- Produces dense, low-dimensional representations
- Prevents memory overflow and accelerates training

### Step 6: Model Training & Evaluation

- Models trained on training split
- Evaluation performed on held-out test set

# 6. Overall Conclusion

- **BoW remains the most effective representation** for this task.
- **MLP achieved the highest overall accuracy**, closely followed by SVM.
- POS and NER features alone are insufficient for document-level classification.
- Combining features offers slightly improvement at best.
- **HMM is fundamentally unsuitable** for non-sequential text representations.